

DRY-RUN GUIDE | quick.py
Source: Dsa\sort\quick.py

CORE BEHAVIOR

Orders values using the comparisons, swaps, insertion, partition, or merges shown in this file.

REPRESENTATIVE INPUT

[4,1,3,2].

STEP TRACE

Each pass/partition places values nearer their final positions; completed sequence is nondecreasing.

EXPECTED OUTPUT / RETURN

1 2 3 4 (check source for exact formatting).

ACTUAL SOURCE EXCERPT

Verbatim excerpt from Dsa\sort\quick.py; line numbers refer to the original file. No source was modified.

```
001 | def quick_sort(arr):  
002 |     if len(arr) <= 1:  
003 |         return arr  
004 |     pivot = arr[len(arr) // 2]  
005 |     left = [x for x in arr if x < pivot]  
006 |     middle = [x for x in arr if x == pivot]  
007 |     right = [x for x in arr if x > pivot]  
008 |     return quick_sort(left) + middle + quick_sort(right)
```

DRY-RUN NOTE

Values are representative. For function-only modules the guide uses an illustrative direct call and explains its return; quirks/errors describe actual behavior.